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Question 1 [15 Points]

Design an Entity-Relationship (ER) diagram for a **Bank Database** that meets the following requirements:

1. The bank employs multiple Employees, and each employee has attributes like Employee_ID, Name (First name, Middle initial, Last name), Date_of_Birth, Address, Gender, and Salary.
2. Each Employee works for one Branch. Each branch has a unique Branch_Code, a unique Branch_Name, and Contact_Numbers (where each branch may have multiple contact numbers).
3. Each branch has one Manager who manages the branch. For each management assignment, the Start_date of the assignment should be recorded. Also, calculate and keep track of the Number of employees in each branch.
4. Employees can be assigned to Projects. Each project has a unique Project_Code, a unique Project_Name, and a Location. For each project an employee works on, record the Hours the employee spends on it.
5. Each Branch can control multiple Projects, but each project is controlled by only one branch.
6. The bank also tracks the Supervisor relationships between employees. Each employee may supervise others and may also be supervised by others.
7. Some employees have Dependents. Each dependent has attributes like Dependent_Name (unique only when combined with Employee_ID), Gender, Birth_date, and Relationship to the employee.



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Question 1 [15 Points]

Design an Entity-Relationship (ER) diagram for a **Telecom Database** for a company that meets the following requirements:

1. The company employs multiple Staff Members, and each staff member has attributes like Staff_ID, Name (First name, Middle initial, Last name), DOB (Date of Birth), Address, Gender, and Monthly_Salary.
2. Each Staff Member works for one Office. Each office has a unique Office_Code, a unique Office_Name, and Contact_Emails (where each office may have multiple contact email addresses).
3. Each office has one Director who manages the office. For each management assignment, the Start_Date of the assignment should be recorded. Also, calculate and keep track of the Total_Employees in each office.
4. Staff Members can be assigned to Tasks. Each task has a unique Task_ID, a unique Task_Name, and an Assigned_Location. For each task a staff member works on, record the Hours_Spent by the staff member.
5. Each Office can control multiple Tasks, but each task is controlled by only one office.
6. The company also tracks the Mentorship relationships between staff members. Each staff member may mentor others and may also be mentored by others.
7. Some staff members have Family Members as dependents. Each family member has attributes like Family_Member_Name (unique only when combined with Staff_ID), Gender, Birth_Date, and Relationship to the staff member.



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Question 1 [15 Points]

Design an Entity-Relationship (ER) diagram for a **Healthcare Database** for an organization called **HealthFirst** that meets the following requirements:

1. The organization hires multiple Healthcare Workers, and each worker has attributes like Worker_ID, Full_Name (First name, Middle initial, Last name), Birth_Date, Home_Address, Gender, and Annual_Salary.
2. Each Healthcare Worker is assigned to a Clinic. Each clinic has a unique Clinic_Code, a unique Clinic_Name, and Contact_Numbers (where each clinic may have multiple contact numbers).
3. Each clinic has one Head_Officer who manages the clinic. For each management assignment, the Start_Date of the assignment should be recorded. Also, calculate and keep track of the Number_of_Workers in each clinic.
4. Healthcare Workers can be assigned to Programs. Each program has a unique Program_ID, a unique Program_Name, and a Program_Location. For each program a worker participates in, record the Hours_Contributed by the worker.
5. Each Clinic can oversee multiple Programs, but each program is supervised by only one clinic.
6. The organization also tracks the Reporting Structure between healthcare workers. Each worker may report to others and may also have others reporting to them.
7. Some healthcare workers have Beneficiaries as dependents. Each beneficiary has attributes like Beneficiary_Name (unique only when combined with Worker_ID), Gender, Birth_Date, and Relationship to the worker.

